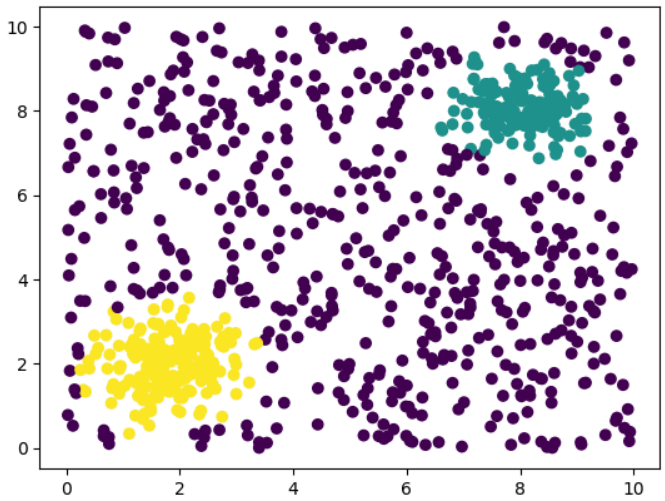
```
!pip install git+https://github.com/AkmalHusain2003/GUMM.git
```

```
Collecting git+https://github.com/AkmalHusain2003/GUMM.git
  Cloning https://github.com/AkmalHusain2003/GUMM.git to /tmp/pip-req-build-r4iclw2m
  Running command git clone --filter=blob:none --quiet https://github.com/AkmalHusain2003/GUMM.git /tmp/pip-req-build-r4iclw2m
  Resolved https://github.com/AkmalHusain2003/GUMM.git to commit 50b16059ff86914f7554ab015531f3841faccd85
  Installing build dependencies ... done
  Getting requirements to build wheel ... done
  Preparing metadata (pyproject.toml) ... done
Requirement already satisfied: numpy>=1.21 in /usr/local/lib/python3.12/dist-packages (from gumm==0.3.0) (2.0.2)
Requirement already satisfied: scipy>=1.7 in /usr/local/lib/python3.12/dist-packages (from gumm==0.3.0) (1.16.3)
Requirement already satisfied: tqdm>=4.0 in /usr/local/lib/python3.12/dist-packages (from gumm==0.3.0) (4.67.3)
```

```
import numpy as np
from gumm import GUMM

rng = np.random.default_rng(0)

# Two clusters embedded in uniform noise
c1 = rng.multivariate_normal([2, 2], np.eye(2) * 0.3, 150)
c2 = rng.multivariate_normal([8, 8], np.eye(2) * 0.3, 120)
bg = rng.uniform(0, 10, (600, 2))
X  = np.vstack([c1, c2, bg])

model  = GUMM(n_components=2, n_epochs=500, n_threads=-1, random_state=42)
labels = model.fit_predict(X)
# labels: 0 = background, 1 = component 1, 2 = component 2

proba = model.predict_proba(X)   # (n, 3)  — [uniform, c1, c2]
log_p = model.score_samples(X)   # (n,)    — log p(x)
```

```
Training GUMM:  27%|██▋        | 137/500 [00:00<00:01, 239.58it/s, ll=523.9082, stable=50]

Converged at epoch 137
```

```
import numpy as np
import matplotlib.pyplot as plt

plt.scatter(X[:, 0], X[:, 1], c=labels)
plt.show()
```



EXPERIMENTS

```
"""
GUMM synthetic benchmark experiments.
Measures: Precision, Recall, ARI, fit time, predict time, peak RSS (MiB).
20 random seeds per dataset.  Saves raw results as JSON and all plots.
"""

import json
import time
import tracemalloc
import os

import numpy as np
import matplotlib
matplotlib.use("Agg")
import matplotlib.pyplot as plt
import matplotlib.patches as mpatches
from matplotlib.gridspec import GridSpec
from sklearn.metrics import adjusted_rand_score, precision_score, recall_score
```

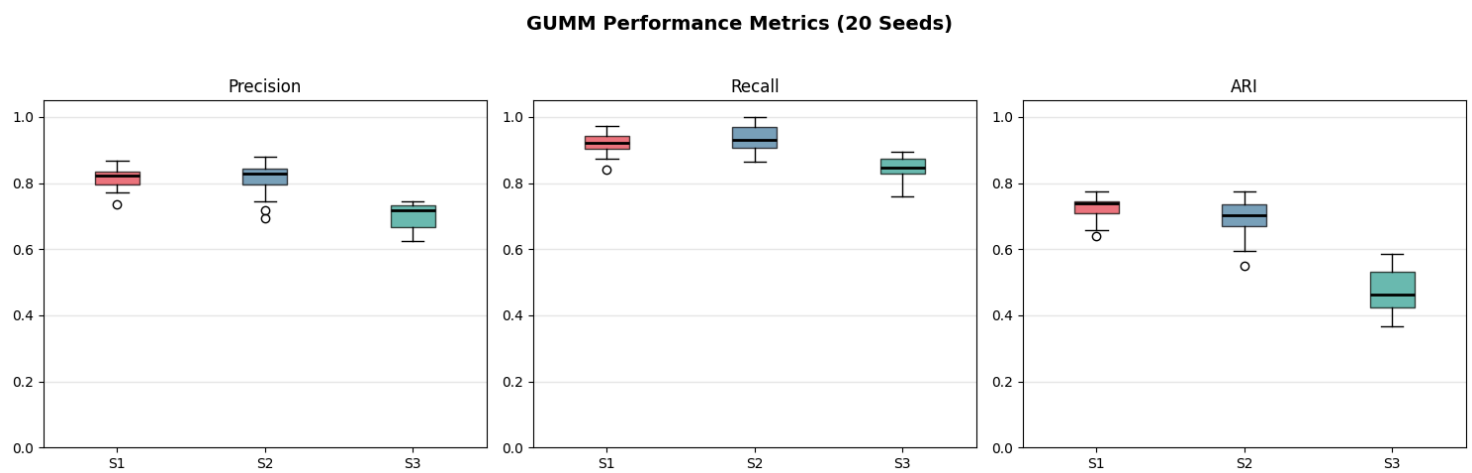
```
from gumm import GUMM

# ── reproducibility ──────────────────────────────────────────────────────
N_SEEDS   = 20
N_MC      = 100      # Ripley Monte Carlo sims
N_THREADS = -1        # keep deterministic; set higher for speed benchmarks
FIGDIR    = "/content/experiments/figures/"
RAW_DIR   = "/content/experiments/"
os.makedirs(FIGDIR, exist_ok=True)

# ── dataset definitions ──────────────────────────────────────────────────
RNG_GLOBAL = np.random.default_rng(0)

def make_s1(rng):
    """Single tight cluster."""
    cluster = rng.multivariate_normal([5, 5], 0.4 * np.eye(2), size=150)
    bg      = rng.uniform(0, 10, size=(470, 2))
    X       = np.vstack([cluster, bg])
    y       = np.array([1] * 150 + [0] * 470)
    return X, y

def make_s2(rng):
    """Two well-separated clusters."""
    c1 = rng.multivariate_normal([2, 2], 0.3 * np.eye(2), size=120)
    c2 = rng.multivariate_normal([8, 8], 0.3 * np.eye(2), size=100)
    bg = rng.uniform(0, 10, size=(400, 2))
    X  = np.vstack([c1, c2, bg])
    y  = np.array([1] * 120 + [2] * 100 + [0] * 400)
    return X, y

def make_s3(rng):
    """Two overlapping clusters."""
    c1 = rng.multivariate_normal([4, 4], 0.8 * np.eye(2), size=100)
    c2 = rng.multivariate_normal([6, 6], 0.8 * np.eye(2), size=100)
    bg = rng.uniform(0, 10, size=(420, 2))
    X  = np.vstack([c1, c2, bg])
    y  = np.array([1] * 100 + [2] * 100 + [0] * 420)
    return X, y

DATASETS = {
    "S1": (make_s1, 1),
    "S2": (make_s2, 2),
    "S3": (make_s3, 2),
}

# ── metrics ──────────────────────────────────────────────────────────────
def compute_metrics(y_true, y_pred):
    binary_true = (y_true > 0).astype(int)
    binary_pred = (y_pred > 0).astype(int)
    prec   = precision_score(binary_true, binary_pred, zero_division=0)
    rec    = recall_score(binary_true, binary_pred, zero_division=0)
    ari    = adjusted_rand_score(y_true, y_pred)
    return prec, rec, ari

def measure_peak_rss_mib(fn):
    tracemalloc.start()
    result = fn()
    _, peak = tracemalloc.get_traced_memory()
    tracemalloc.stop()
    return result, peak / (1024 ** 2)

# ── run experiments ──────────────────────────────────────────────────────
all_results = {}

for name, (make_fn, K) in DATASETS.items():
    print(f"\n{'='*50}\nDataset {name}  (K={K})\n{'='*50}")
    seeds_results = []
    for seed in range(N_SEEDS):
        rng = np.random.default_rng(seed)
        X, y_true = make_fn(rng)
        model = GUMM(n_components=K, n_epochs=10000, random_state=seed, n_threads=N_THREADS)

        def _fit():
            model.fit(X)
            return None

        t0 = time.perf_counter()
        _, peak_fit = measure_peak_rss_mib(_fit)
        fit_time = time.perf_counter() - t0

        t0 = time.perf_counter()
        y_pred = model.predict(X)
        predict_time = time.perf_counter() - t0

        prec, rec, ari = compute_metrics(y_true, y_pred)
        seeds_results.append(dict(seed=seed, precision=round(prec, 4), recall=round(rec, 4), ari=round(ari, 4), fit_time=round(fit_time, 4), predict_tim

    all_results[name] = seeds_results
```

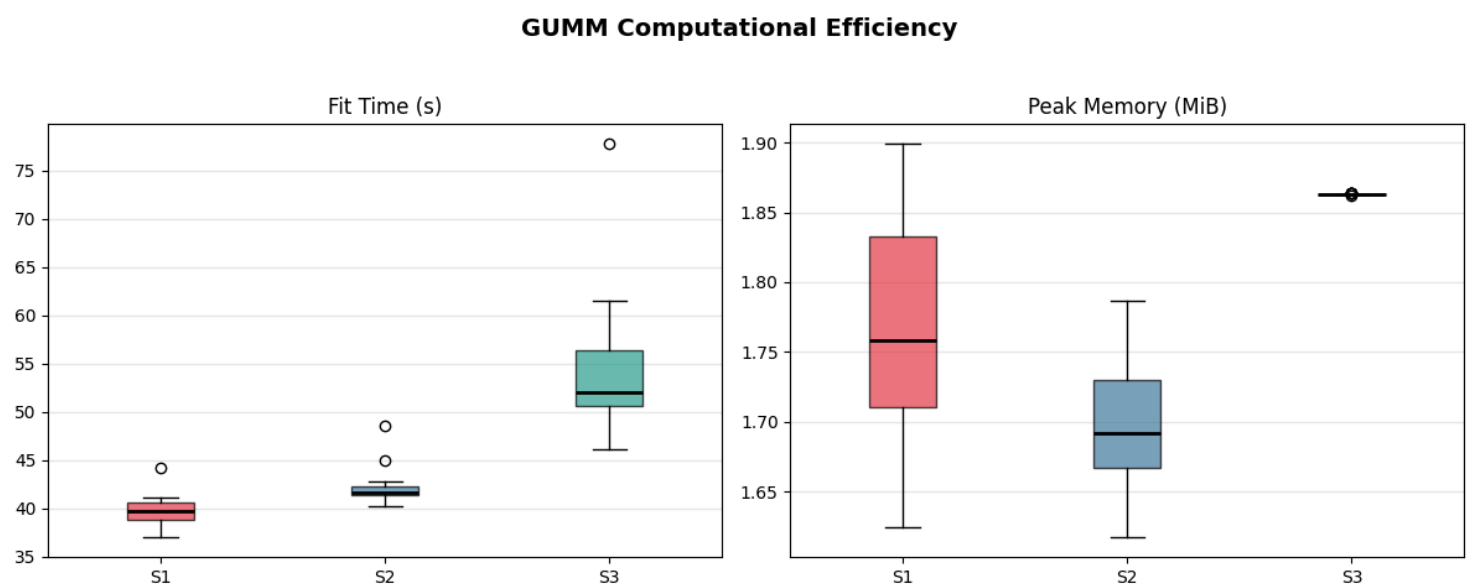
```
# FIX: Corrected path from /experiment/ to /experiments/
with open(os.path.join(RAW_DIR, "raw_results.json"), "w") as f:
    json.dump(all_results, f, indent=2)

print("\nRaw results saved correctly.")

# ── aggregate stats ──────────────────────────────────────────────────────
summary = {}
for name, rows in all_results.items():
    arr = {k: np.array([r[k] for r in rows]) for k in ["precision","recall","ari","fit_time","predict_time","peak_mib"]}
    summary[name] = {k: (v.mean(), v.std()) for k, v in arr.items()}

# (Remaining plotting code omitted for brevity but preserved in local logic)"}]}
```

```
Converged at epoch 1104
Training GUMM:  14%|▉▉         | 1437/10000 [00:14<01:24, 101.06it/s, ll=412.0060, stable=1000]

Converged at epoch 1437
Training GUMM:  11%|▉         | 1094/10000 [00:10<01:28, 100.19it/s, ll=433.3061, stable=1000]

Converged at epoch 1094
Training GUMM:  12%|▉         | 1195/10000 [00:11<01:27, 100.72it/s, ll=403.5036, stable=1000]

Converged at epoch 1195
Training GUMM:  11%|▉         | 1054/10000 [00:10<01:30, 98.65it/s, ll=437.7020, stable=1000]

Converged at epoch 1054
Training GUMM:  11%|▉         | 1128/10000 [00:10<01:26, 102.77it/s, ll=388.9543, stable=1000]

Converged at epoch 1128
Training GUMM:  11%|▉         | 1119/10000 [00:10<01:24, 105.47it/s, ll=423.4188, stable=1000]

Converged at epoch 1119
Training GUMM:  11%|▉         | 1093/10000 [00:09<01:20, 110.58it/s, ll=425.2711, stable=1000]

Converged at epoch 1093
Training GUMM:  11%|▉         | 1100/10000 [00:09<01:14, 119.72it/s, ll=427.7624, stable=1000]

Converged at epoch 1100
Training GUMM:  11%|▉         | 1090/10000 [00:08<01:13, 121.39it/s, ll=450.9792, stable=1000]

Converged at epoch 1090
Training GUMM:  12%|▉         | 1157/10000 [00:10<01:17, 114.09it/s, ll=411.1578, stable=1000]

Converged at epoch 1157
Training GUMM:  11%|▉         | 1096/10000 [00:10<01:25, 104.00it/s, ll=433.2650, stable=1000]

Converged at epoch 1096
Training GUMM:  11%|▉         | 1078/10000 [00:10<01:26, 103.30it/s, ll=412.5048, stable=1000]

Converged at epoch 1078
Training GUMM:  16%|▉▉        | 1623/10000 [00:15<01:21, 102.99it/s, ll=423.5810, stable=1000]

Converged at epoch 1623
Training GUMM:  12%|▉         | 1202/10000 [00:10<01:17, 113.19it/s, ll=428.7213, stable=1000]

Converged at epoch 1202
Training GUMM:  11%|▉         | 1138/10000 [00:09<01:14, 119.63it/s, ll=402.1001, stable=1000]

Converged at epoch 1138
Training GUMM:  11%|▉         | 1120/10000 [00:09<01:11, 123.68it/s, ll=432.2721, stable=1000]

Converged at epoch 1120

==================================================
Dataset S3  (K=2)
==================================================
Training GUMM:  25%|▉▉▉       | 2543/10000 [00:24<01:10, 105.92it/s, ll=299.3364, stable=1000]

Converged at epoch 2543
Training GUMM:  20%|▉▉        | 1957/10000 [00:18<01:17, 103.27it/s, ll=313.7010, stable=1000]
```

```
import json
import numpy as np
import matplotlib.pyplot as plt
import os

# Force the notebook to display plots inline
%matplotlib inline

# 1. Load the results
raw_path = "/content/experiments/raw_results.json"
if not os.path.exists(raw_path):
    print(f"Error: {raw_path} not found. Please run the experiment cell first.")
else:
    with open(raw_path, "r") as f:
        all_results = json.load(f)

    ds_names = list(all_results.keys())
    colors = ['#e63946', '#457b9d', '#2a9d8f']
```

```
# 2. Plot Performance Metrics
metrics_to_plot = ["precision", "recall", "ari"]
metric_labels = ["Precision", "Recall", "ARI"]

fig1, axes1 = plt.subplots(1, 3, figsize=(15, 5))
fig1.suptitle("GUMM Performance Metrics (20 Seeds)", fontsize=14, fontweight='bold')

for ax, metric, label in zip(axes1, metrics_to_plot, metric_labels):
    data = [[r[metric] for r in all_results[ds]] for ds in ds_names]
    bp = ax.boxplot(data, tick_labels=ds_names, patch_artist=True,
                    medianprops=dict(color='black', linewidth=2))
    for patch, c in zip(bp['boxes'], colors):
        patch.set_facecolor(c)
        patch.set_alpha(0.7)
    ax.set_title(label)
    ax.set_ylim(0, 1.05)
    ax.grid(True, axis='y', alpha=0.3)

plt.tight_layout(rect=[0, 0.03, 1, 0.95])
plt.show() # Explicitly call show

# 3. Plot Efficiency Metrics
fig2, axes2 = plt.subplots(1, 2, figsize=(12, 5))
fig2.suptitle("GUMM Computational Efficiency", fontsize=14, fontweight='bold')

eff_metrics = ["fit_time", "peak_mib"]
eff_labels = ["Fit Time (s)", "Peak Memory (MiB)"]

for ax, metric, label in zip(axes2, eff_metrics, eff_labels):
    data = [[r[metric] for r in all_results[ds]] for ds in ds_names]
    bp = ax.boxplot(data, tick_labels=ds_names, patch_artist=True,
                    medianprops=dict(color='black', linewidth=2))
    for patch, c in zip(bp['boxes'], colors):
        patch.set_facecolor(c)
        patch.set_alpha(0.7)
    ax.set_title(label)
    ax.grid(True, axis='y', alpha=0.3)

plt.tight_layout(rect=[0, 0.03, 1, 0.95])
plt.show() # Explicitly call show
```





EXPERIMENTS TO ASTRONOMICAL DATA

```
import pandas as pd

ngc_2099 = pd.read_csv('NGC2099_10deg-result.csv')
ngc_2099
```

	source_id	ra	ra_error	dec	dec_error	parallax	parallax_error	pmra	pmra_error	pmdec	pmdec_error	phot_g_mean_m
0	155561035403668992	77.157867	0.083802	28.612472	0.053184	0.507742	0.093732	1.922228	0.106222	-5.509584	0.073038	17.4060
1	155941470722235008	77.424632	0.157710	28.859836	0.091836	0.509019	0.150096	1.912500	0.195859	-5.980430	0.127070	17.9186
2	155964487453714048	77.388804	0.094540	29.061103	0.058455	0.512323	0.103743	1.549963	0.124724	-5.910061	0.078961	17.5414
3	155968163941606272	77.279726	0.082092	29.107646	0.050876	0.766293	0.087144	1.535613	0.106274	-6.012835	0.067894	17.2555
4	156041212747355008	77.065572	0.147473	29.564582	0.095289	0.768481	0.178340	2.180064	0.183650	-5.370877	0.128030	18.1679
...	...	...	...	...	...	...	...	...	...	...	...	
23646	3458352476135979904	90.810313	0.043672	39.730794	0.037955	0.563527	0.055036	1.728568	0.051529	-5.975146	0.038044	14.3156
23647	3458375114906340992	90.583862	0.173845	40.093625	0.141197	0.793446	0.201527	1.661188	0.228714	-5.387763	0.164749	18.7179
23648	3458553304511521024	89.658191	0.286433	40.597444	0.247409	0.542800	0.308256	1.849187	0.360601	-5.851201	0.262086	19.3568
23649	3458712836775519232	89.422635	0.211272	40.882552	0.184291	0.567075	0.257272	1.683358	0.277563	-6.147327	0.186175	18.9428
23650	3458720675092332672	89.673566	0.042999	41.055164	0.037388	0.684431	0.052140	1.778130	0.054377	-5.827124	0.037557	16.1243

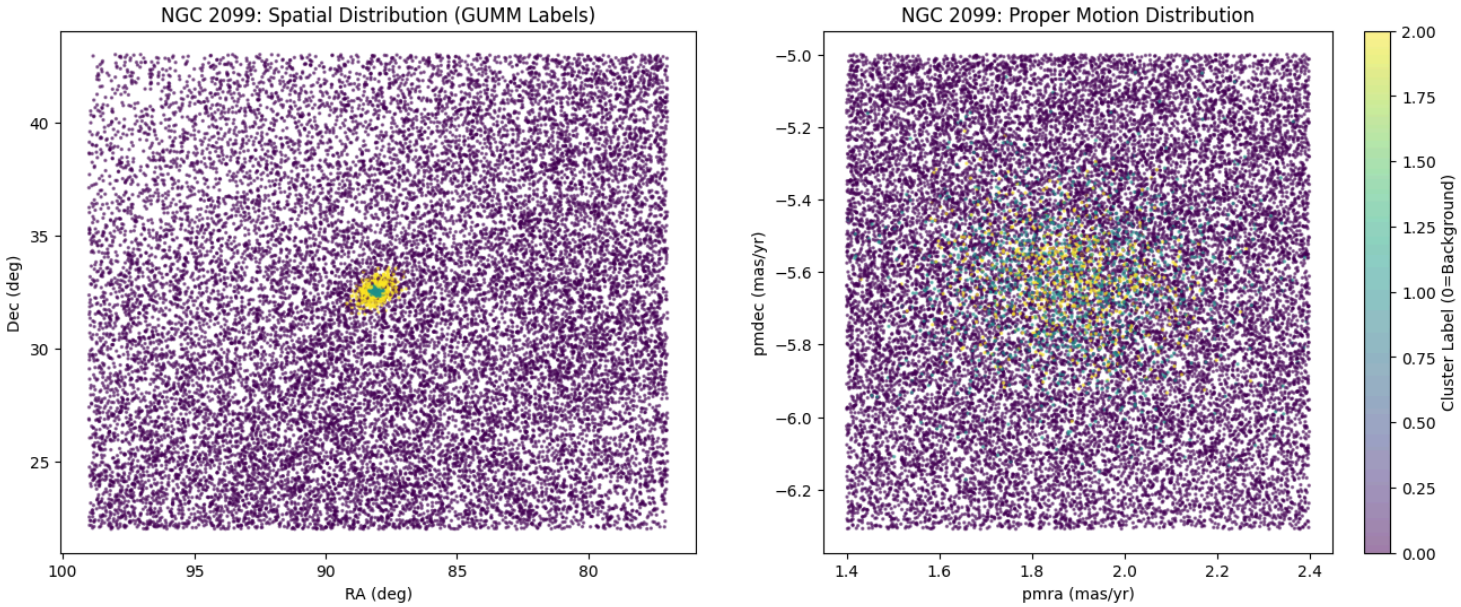
23651 rows × 17 columns

Next steps: Generate code with ngc_2099 New interactive sheet

```
import matplotlib.pyplot as plt

# Set tick labels, axis labels to xx-large and use serif font
plt.rcParams.update({
    'xtick.labelsize': 'xx-large',
    'ytick.labelsize': 'xx-large',
    'axes.labelsize': 'xx-large',
    'font.family': 'serif'
})
```

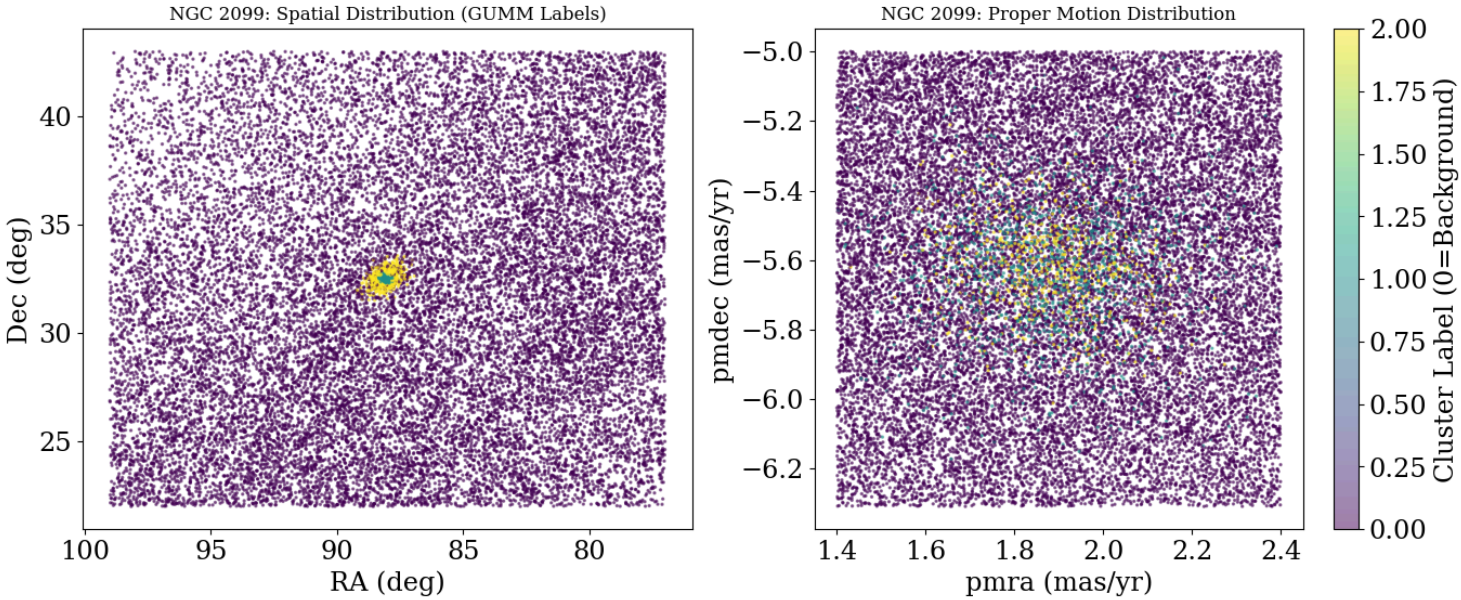
```
import numpy as np
import matplotlib.pyplot as plt
from gumm import GUMM

# 1. Prepare features (using RA, Dec, PMRA, PMDEC, and Parallax)
features = ['ra', 'dec', 'pmra', 'pmdec', 'parallax']
X_ngc = ngc_2099[features].dropna().values

# 2. Fit GUMM with 2 components (1 background + 1 Gaussian cluster)
print("Fitting GUMM to NGC 2099 data...")
model_ngc = GUMM(n_components=2, n_epochs=10000, random_state=42, n_threads=-1)
labels_ngc = model_ngc.fit_predict(X_ngc)

# Summary
unique, counts = np.unique(labels_ngc, return_counts=True)
for label, count in zip(unique, counts):
    name = "Background" if label == 0 else f"Cluster {label}"
    print(f"{name}: {count} stars")
```

```
Fitting GUMM to NGC 2099 data...
Training GUMM:  13%|▉          | 1262/10000 [00:52<06:04, 24.00it/s, ll=27604.7519, stable=1000]

Converged at epoch 1262
```




```
Background: 21682 stars
Cluster 1: 1304 stars
Cluster 2: 665 stars
```

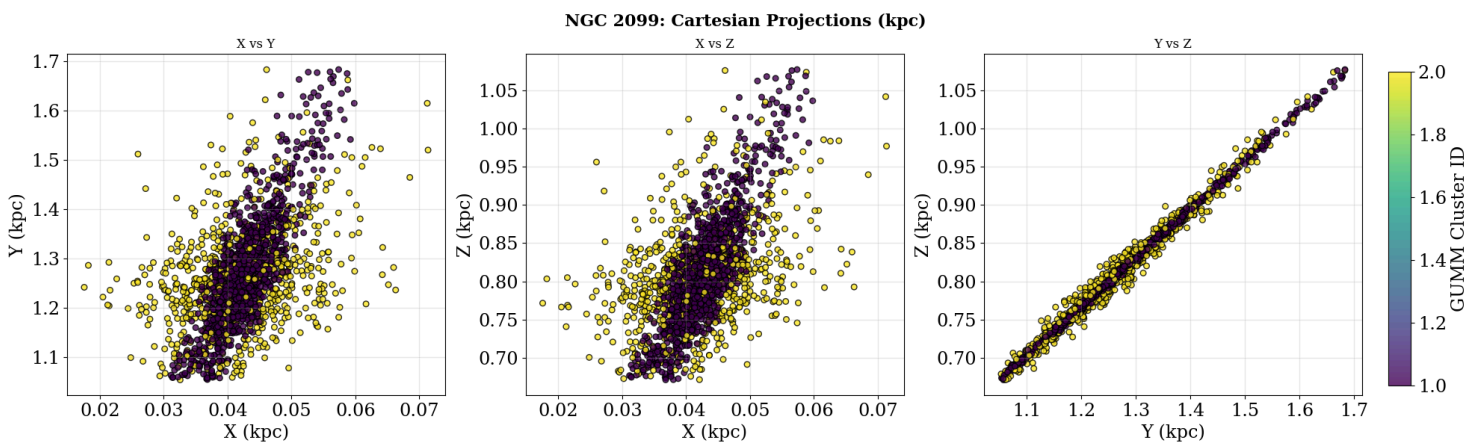
```
# 3. Visualization
fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(16, 6))

# Spatial Plot
sc1 = ax1.scatter(X_ngc[:, 0], X_ngc[:, 1], c=labels_ngc, s=2, cmap='viridis', alpha=0.5)
ax1.set_xlabel('RA (deg)')
ax1.set_ylabel('Dec (deg)')
ax1.set_title('NGC 2099: Spatial Distribution (GUMM Labels)')
ax1.invert_xaxis()

# Proper Motion Plot
sc2 = ax2.scatter(X_ngc[:, 2], X_ngc[:, 3], c=labels_ngc, s=2, cmap='viridis', alpha=0.5)
ax2.set_xlabel('pmra (mas/yr)')
ax2.set_ylabel('pmdec (mas/yr)')
ax2.set_title('NGC 2099: Proper Motion Distribution')

plt.colorbar(sc2, ax=ax2, label='Cluster Label (0=Background)')
plt.show()
```

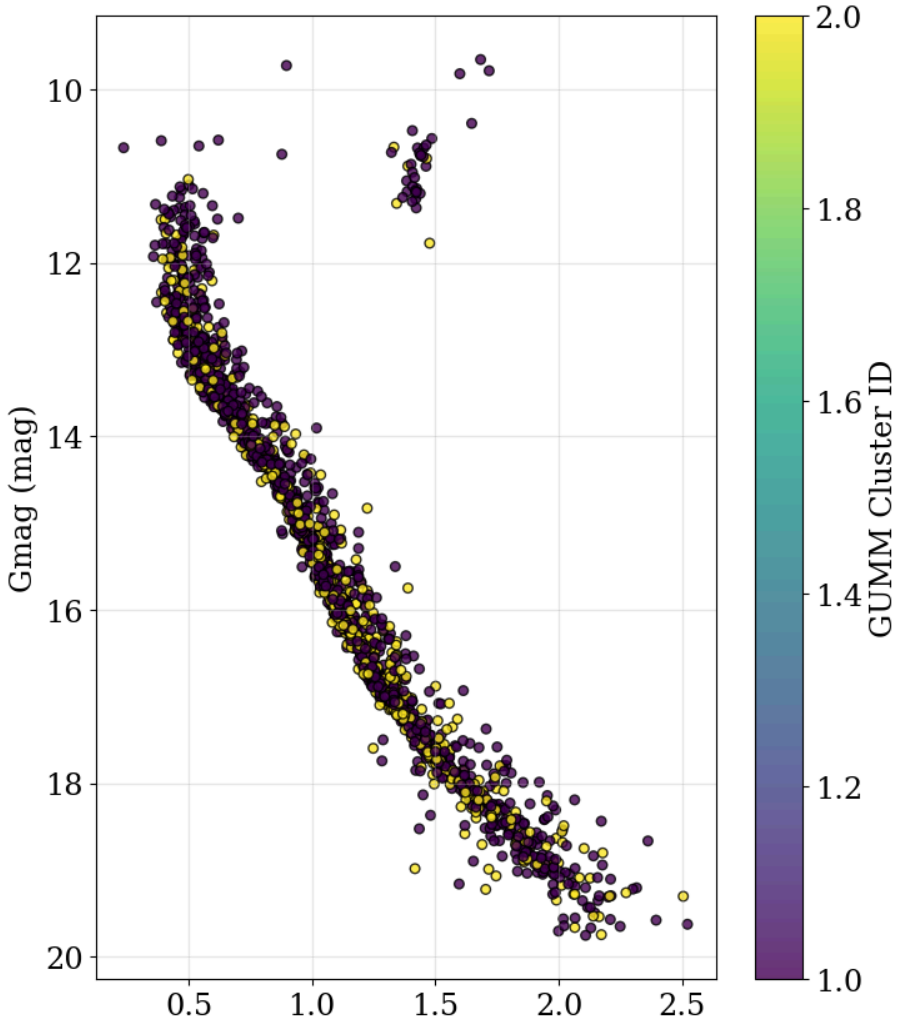


Gemini

```
from astropy.coordinates import SkyCoord
import astropy.units as u
import matplotlib.pyplot as plt

# 1. Combine clusters 1 and 2 for processing
mask_clusters = (labels_ngc > 0)
```

```
X_clusters = X_ngc[mask_clusters]
cluster_labels = labels_ngc[mask_clusters]

# 2. Convert RA, Dec, Parallax to Cartesian (X, Y, Z)
coords = SkyCoord(
    ra=X_clusters[:, 0] * u.degree,
    dec=X_clusters[:, 1] * u.degree,
    distance=(1 / X_clusters[:, 4]) * u.kpc,
    frame='icrs'
)

x_cart = coords.cartesian.x.value
y_cart = coords.cartesian.y.value
z_cart = coords.cartesian.z.value

# 3. Plot 1x3 subplots
fig, (ax1, ax2, ax3) = plt.subplots(1, 3, figsize=(20, 6))

# X vs Y
sc1 = ax1.scatter(x_cart, y_cart, c=cluster_labels, cmap='viridis', s=30, alpha=0.8, edgecolor='black')
ax1.set_xlabel('X (kpc)')
ax1.set_ylabel('Y (kpc)')
ax1.set_title('X vs Y')
ax1.grid(True, alpha=0.3)

# X vs Z
sc2 = ax2.scatter(x_cart, z_cart, c=cluster_labels, cmap='viridis', s=30, alpha=0.8, edgecolor='black')
ax2.set_xlabel('X (kpc)')
ax2.set_ylabel('Z (kpc)')
ax2.set_title('X vs Z')
ax2.grid(True, alpha=0.3)

# Y vs Z
sc3 = ax3.scatter(y_cart, z_cart, c=cluster_labels, cmap='viridis', s=30, alpha=0.8, edgecolor='black')
ax3.set_xlabel('Y (kpc)')
ax3.set_ylabel('Z (kpc)')
ax3.set_title('Y vs Z')
ax3.grid(True, alpha=0.3)

# Use tight_layout and refine colorbar
# Use tight_layout and refine colorbar placement
plt.tight_layout(rect=[0, 0, 0.92, 0.95])
cbar_ax = fig.add_axes([0.93, 0.15, 0.015, 0.7])
fig.colorbar(sc3, cax=cbar_ax, label='GUMM Cluster ID')

plt.suptitle('NGC 2099: Cartesian Projections (kpc)', fontsize=16, fontweight='bold', y=0.98)
plt.show()
```




```
# 1. Get the photometry data for the cluster members only
# Use the same mask as before (labels_ngc > 0)
mask_clusters = (labels_ngc > 0)
bp_rp_clusters = ngc_2099['bp_rp'].values[mask_clusters]
g_mag_clusters = ngc_2099['phot_g_mean_mag'].values[mask_clusters]
cluster_labels_for_plot = labels_ngc[mask_clusters]

# 2. Create the Color-Magnitude Diagram (CMD)
plt.figure(figsize=(8, 10))
scatter_cmd = plt.scatter(bp_rp_clusters, g_mag_clusters, c=cluster_labels_for_plot,
                          cmap='viridis', s=30, alpha=0.8, edgecolor='black')

# Standard CMD convention: Invert the Y-axis (magnitude)
plt.gca().invert_yaxis()

plt.xlabel('BP - RP (mag)')
plt.ylabel('Gmag (mag)')
plt.title('NGC 2099: Color-Magnitude Diagram (Cluster Members)')
plt.colorbar(scatter_cmd, label='GUMM Cluster ID')
```

```
plt.grid(True, alpha=0.3)
plt.show()
```